

Data Structures and Algorithm

COURSE CODE: CS-201

TOPIC: LINKED LIST/ SINGLE LINK LIST

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Goals for today

To study:

- Linked list
 - Singly/ one-way linked list

CLO Covered

CLO1: **Explain** the fundamentals of data structure as well as algorithms that operate on them

Problem With Array

Fixed Length (Occupy a Block of memory) [**Are also called Dense List**]

To expand an Array

- create a new array, longer in size and
- copy the contents of the old array into the new array

Insertion or Deletion

Solution

Attach a pointer to each item in the array, which points to the next item

- This is a *linked list*
- A data item plus its pointer is called a ***node***

Linked List

A linked list, or one-way list, is a linear collection of data elements, called **nodes**, where the linear order is given by means of **pointer**

Linked List

- A linked list is a linear data structure, in which the elements are not stored at contiguous memory locations.
- The elements in a linked list are linked using pointers (entity that points to the next element)

Linked List

- A linked list is a series of connected *nodes*, where each node is a data structure. A linked list can grow or shrink in size as the program runs.
- If new data need to be added to a linked list, the program simply allocates another node and inserts it into the series.
- If a particular piece of data needs to be removed from the linked list, the program deletes the node containing that data.

Linked List

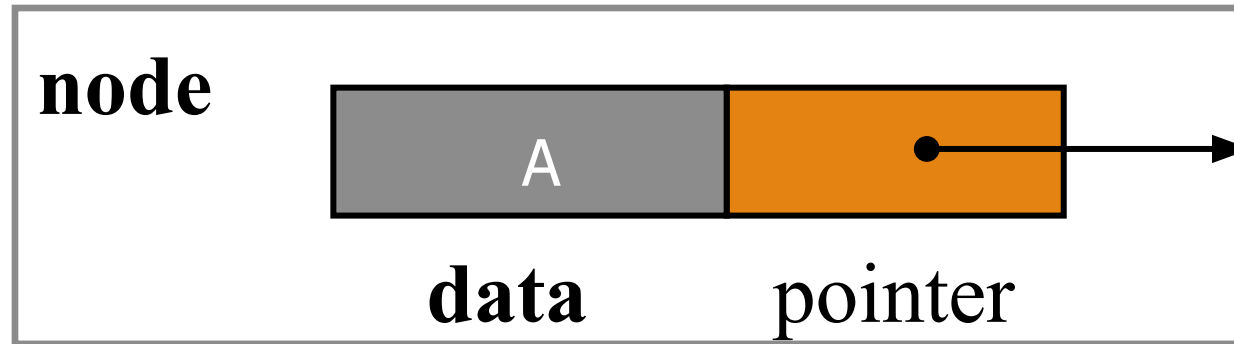
Linked list consists of nodes where each node contains a data field and a reference(link) to the next node in the list.

Each node is divided into two parts

- First part contains the **information** of the element, and
- Second part called the **link field** or **next pointer field** contains the address of the next node in the list.

Linked list is the second most-used data structure after array.

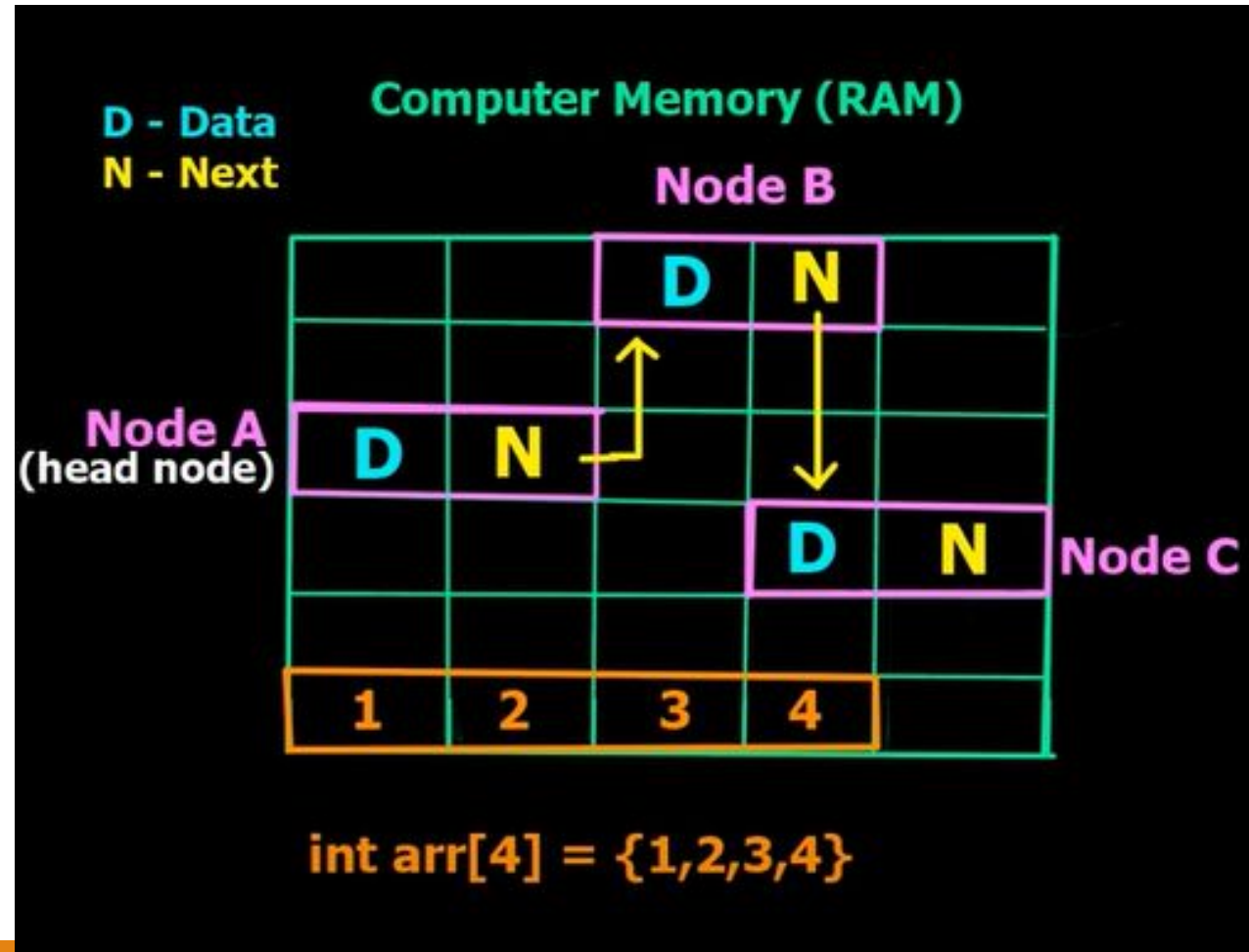
Linked Lists



Each node contains at least

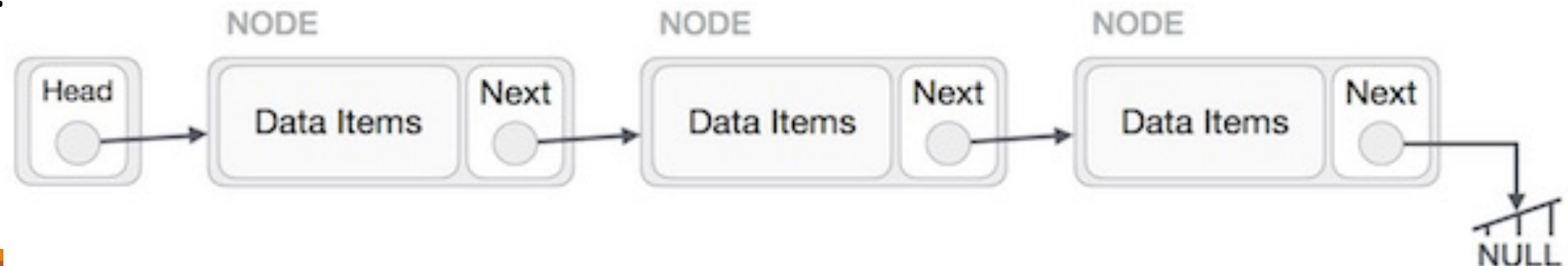
- A piece of data (any type)
- Pointer to the next node in the list

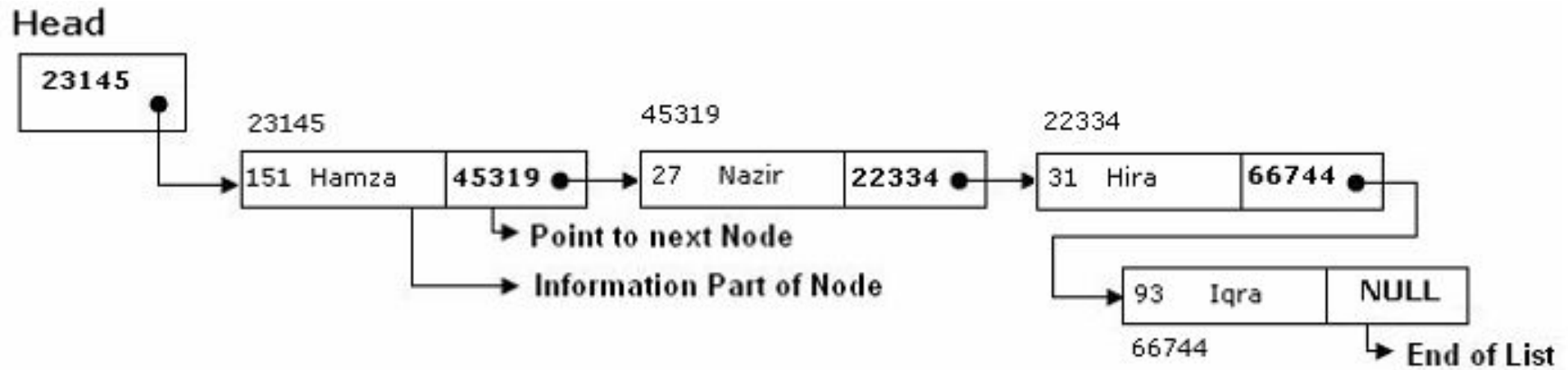
Linked List Representation in Memory



The Composition of Linked Lists

- The following list has nodes, plus a pointer known as the list head.
- The head simply points to the first node in the list. Each node, in turn, points to the next node in the list.
- The first node points to the second node, which points to the third node. Because the third node is the last one in the list, it points to the NULL address (address 0).
- This is usually how the end of a linked list is signified by letting the last node point to NULL.





The Head is a special pointer variable which contains the address of the first node of the list. If there is no node available in the list then Head contains NULL value that means, List is empty.

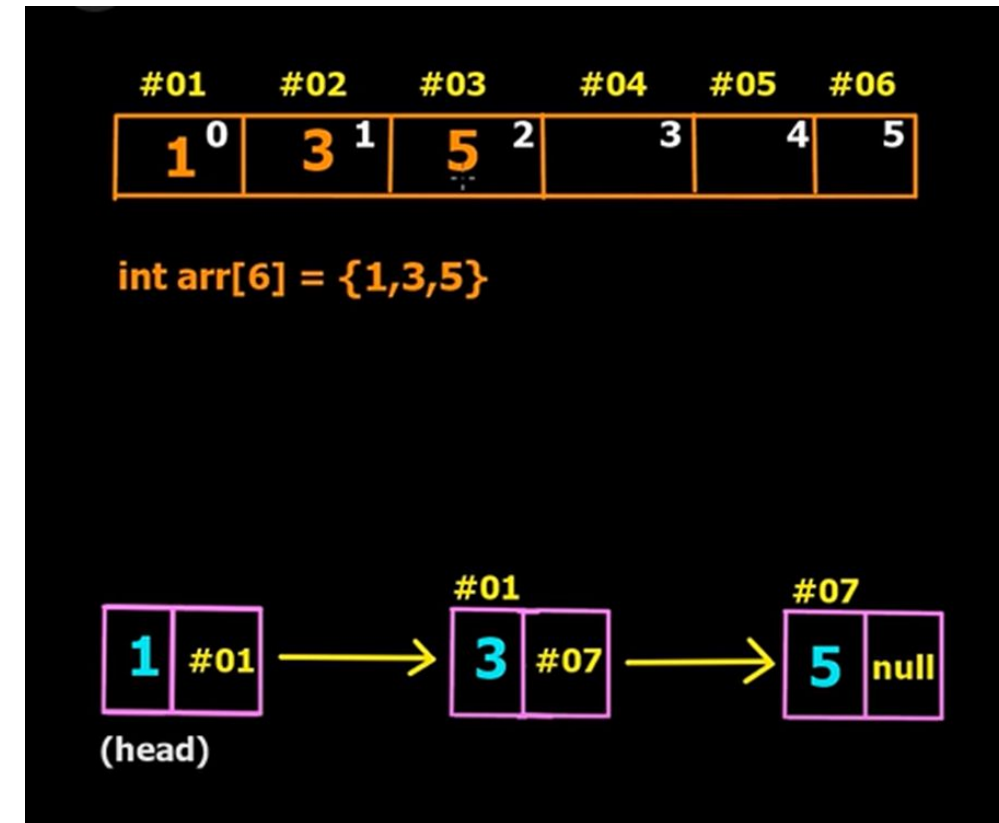
The left part of the each node represents the information part of the node, which may contain an entire record of data (e.g. ID, name, marks, age etc). the right part represents pointer/link to the next node.

The next pointer of the last node is null pointer signal the end of the list

Advantages:

List of data can be stored in arrays but linked structures (pointers) provide several advantages.

- A linked list is appropriate when the number of data elements to be represented in data structure is unpredictable. It also appropriate when there are frequently insertions & deletions occurred in the list.
- Linked lists are dynamic, so the length of a list can increase or decrease as necessary.



Disadvantages

- Random access is not allowed. We have to access elements sequentially starting from the first node.
- Extra memory space for a pointer is required with each element of the list.
- Not cache friendly. Since array elements are contiguous locations, there is locality of reference which is not there in case of linked list

Applications of Linked list

Image viewer

Navigation in web browser

Music player

No. of users visiting a website

Web browser history

Types of Linked List

- Following are the various types of linked list.
 - **Singly Linked List** – Item navigation is forward only.
 - **Doubly Linked List** – Items can be navigated forward and backward.
 - **Circular Linked List** – Last item contains link of the first element as next and the first element has a link to the last element as previous.